Ch 9.3-4: Support Vector Machine

Lecture 24 - CMSE 381

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Weds, Apr 13, 2022

Announcements

Last time:

- Maximal Margin Classifier
- Support Vector Classifier

This lecture:

- Support Vector Machine

Announcements:

- Mon 4/11: 9.1-2
- Weds 4/13: 9.3-4
- Fri 4/15: Lab: 9.6
  - HW #9 Due Friday
- Week of 4/18: Ch 10 Neural nets
- Week of 4/25: Ch 12 Clustering
- Final: May 4, 5:45 - 7:45 PM, Erickson 226

Weds

Section 1

Last Time

Classification Setup

Data matrix:

$$X = \begin{pmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{pmatrix}_{n \times p}$$

$$x_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \cdots, x_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

Observations in one of two classes,
$y_i \in \{-1, 1\}$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Separate out a test observation

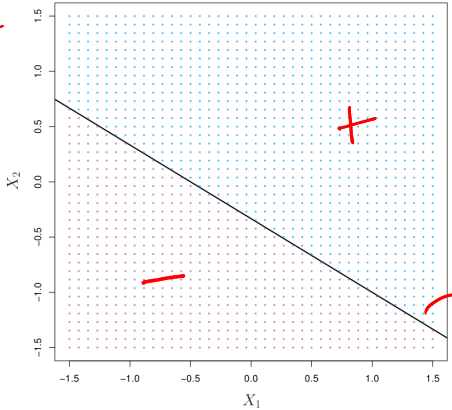
$$x^* = (x_1^* \ \cdots \ x_p^*)^T$$

There are two sides to every hyperplane

In $\mathbb{R}^p$

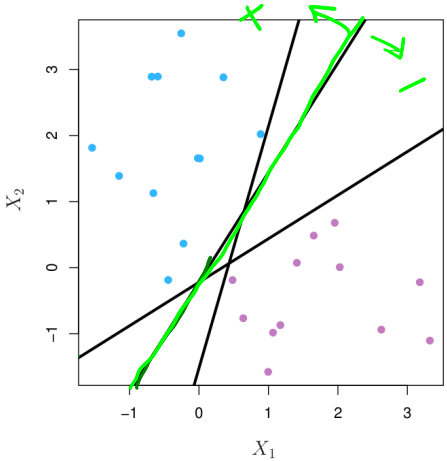
$$\underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p < 0}_{\text{hyperplane}} \qquad \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p > 0$$

X for which this is 0



$1 + 2x_1 + 3x_2 = 0$

Separating Hyperplane



$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$

-or-

For all i:

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) > 0$$

Separating hyperplane becomes a classifier

If you have a separating hyperplane:

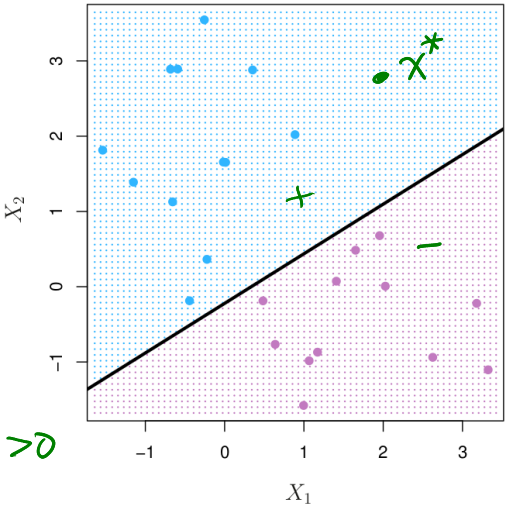
- Check

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

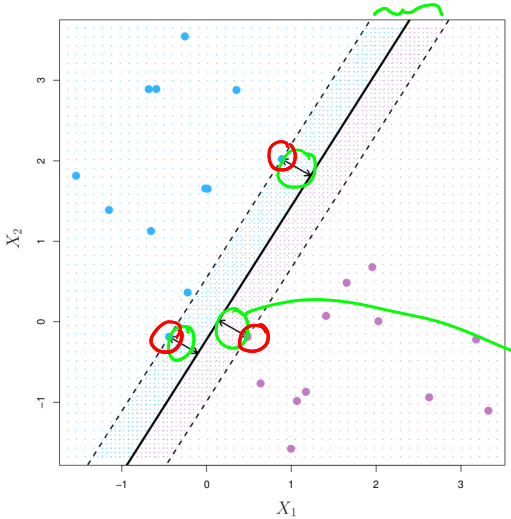
- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$

ex)

$f(x^*) > 0$



How do we pick? Maximal margin classifier



Maximal margin hyperplane is the one w/ largest margin

→ distance from hyperplane to closest data pt

dist = margin

Pts at distance = margin called support vectors

Mathematical Formulation

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M$$

distance margin

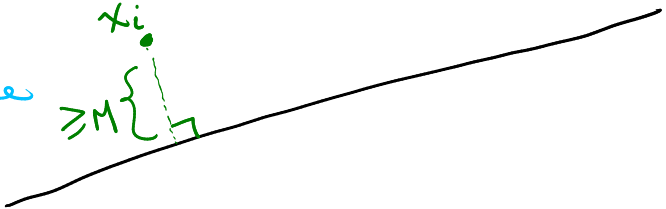
$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

unit length normal

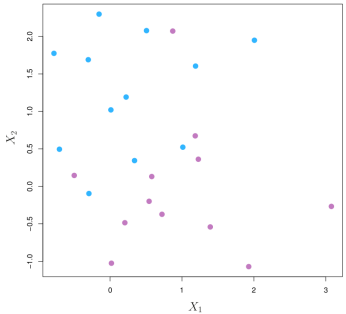
$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \quad \forall\, i = 1, \ldots, n$$

every data pt

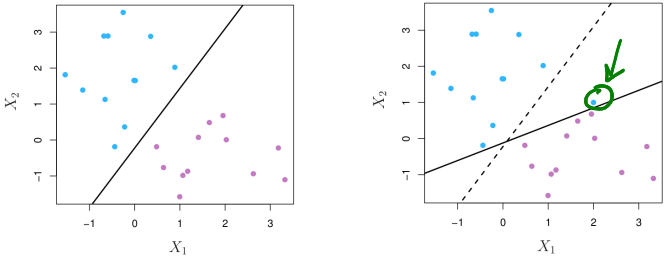
dist from x_i to hyperplane



Issues

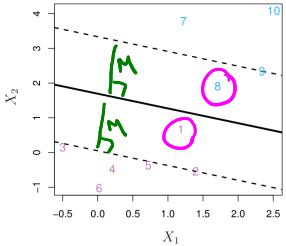


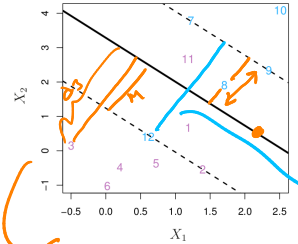
No separating hyperplane exists



Choice of hyperplane is sensitive to new points

Support Vector Classifier





$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

hyper parameter

$\varepsilon_{12} > 1$

$\varepsilon_3 = 0$ $\quad 0 \leq \varepsilon_8 \leq 1$

If $\varepsilon = 0$ $\quad M$

If $\varepsilon = 1$ $\quad M(1-1) = 0$

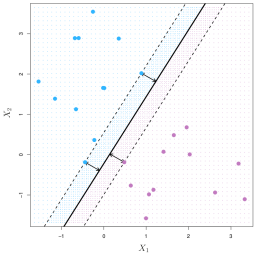
Two formulations side by side

Maximal Margin Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \quad \forall\, i = 1, \ldots, n$$



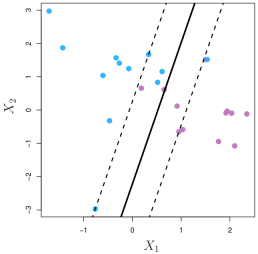
Support Vector Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$



So many variables

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1-\epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

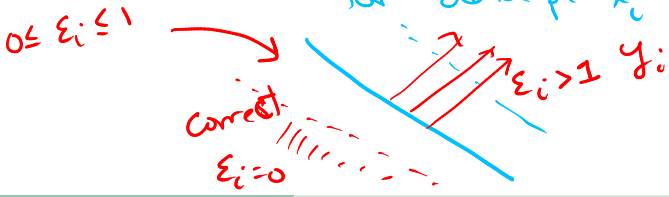
C : non-negative tuning parameter

controls $\sum \varepsilon_i$

"incorrectness"

M : width of margin

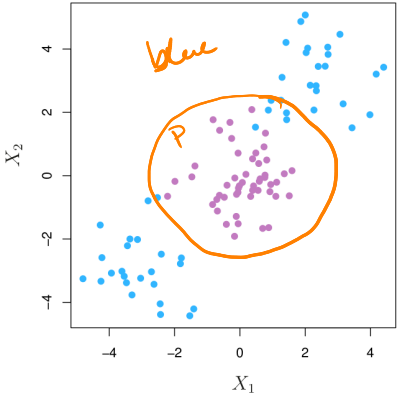
ε_i's : measure of incorrectness for data pt x_i

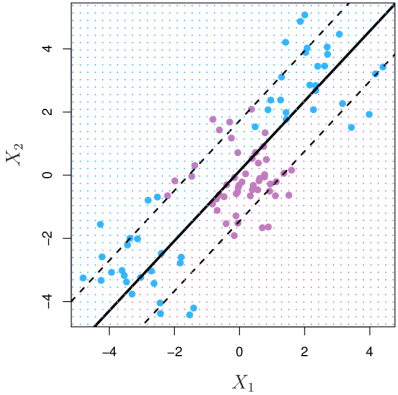


Section 2

Support Vector Machine

Limiting factor of SVC





- Require linear boundaries
- Non-linear options are many

Example of using more features

Want $2p$ features:

$$X_1, X_1^2, X_2, X_2^2, \cdots, X_p, X_p^2$$

Optimization becomes:

SVC

$$\underset{\beta_0,\beta_{11},\beta_{12},\ldots,\beta_{p1},\beta_{p2},\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to } y_i\left(\beta_0 + \sum_{j=1}^{p}\beta_{j1}x_{ij} + \sum_{j=1}^{p}\beta_{j2}x_{ij}^2\right) \geq M(1-\epsilon_i),$$

$$\sum_{i=1}^{n}\epsilon_i \leq C, \quad \epsilon_i \geq 0, \quad \sum_{j=1}^{p}\sum_{k=1}^{2}\beta_{jk}^2 = 1.$$

Sum of all non intercept coeff.

Kernels

Idea: Enlarge feature space

Make sure its computationally efficient

You can pick one

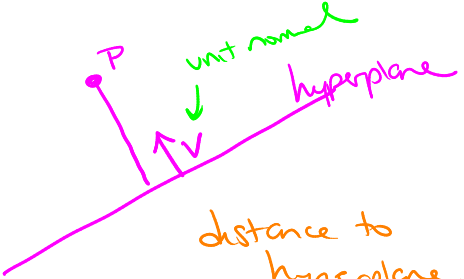
Inner products

$$\langle a, b \rangle = \sum_{i=1}^{r} a_i b_i$$

$(a_1 \; a_2 \; a_3 \; \ldots \; a_r)$

$(b_1 \; b_2 \; b_3 \; \ldots \; b_r)$

$a_1 b_1 + a_2 b_2 + \ldots + a_r b_r$



distance to hyperplane

$\beta_0 + \langle p, v \rangle$

SVC via inner products

Before

$$f(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

$$= \beta_0 + \langle \beta, x \rangle$$

$\Downarrow$ algebra (not shown)

Find coeff

$$f(x) = \beta_0 + \sum_{i=1}^{n} \alpha_i \langle x, x_i \rangle$$

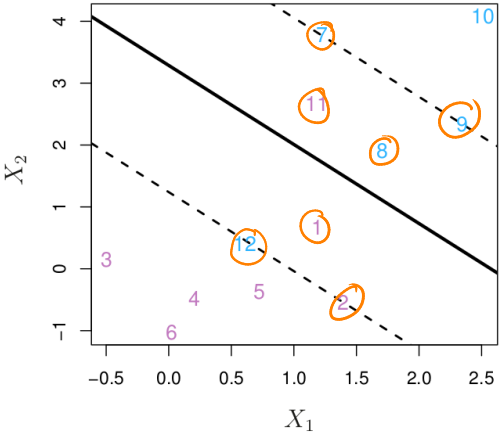
Estimate $\binom{n}{2}$ parameters with respect to $\langle x_i, x_j \rangle$

SVC via inner products of support vectors

Turns out α_i only non-zero for support vectors

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i \langle x, x_i \rangle$$

only need support vectors



Fewer terms $\Rightarrow$ fewer computations for $f(x)$

The kernel

$$K(x_i, x_i')$$

Eats 2 pts in $\mathbb{R}^d$

returns number in $\mathbb{R}$

$$\langle x, x' \rangle = \sum_{i=1}^{d} x_i x_i'$$

called a linear kernel

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

Use SVC setup

with some non-linear

kernel

Result:

Support Vector Machine

A polynomial kernel

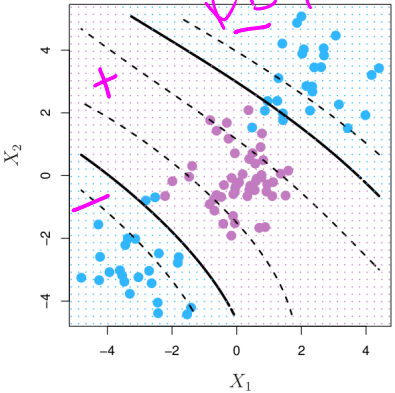
$$\begin{pmatrix} - & x_i & - \\ - & x_{i'} & - \end{pmatrix} X$$

$$K(x_i, x_{i'}) = \left(1 + \overbrace{\sum_{j=1}^{p} x_{ij} x_{i'j}}^{\langle x_i, x_{i'} \rangle}\right)^d \leftarrow \text{degree}$$

$\beta_0 + \sum_{i \in S} \alpha_i K(x, x_i)$

Degree $d = 3$

Result: More flexible decision boundaries



A radial kernel

dist $\sqrt{\sum (x_{ij} - x_{i'j})^2}$

$\gamma > 0$ positive tuning parameter

x_i

x_i'

$$K(x_i, x_i') = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right)$$

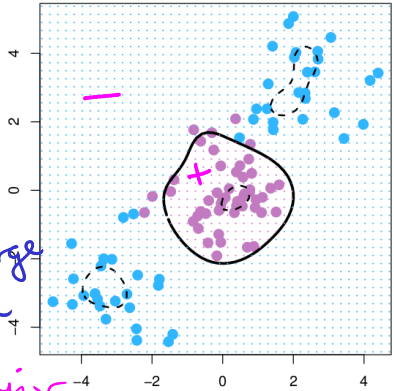
If x_i, x_i' nearby $\rightsquigarrow$ summation is small

$e^{-\text{small}} \Rightarrow$ big #

If x_i, x_i' far $\rightsquigarrow$ summation large

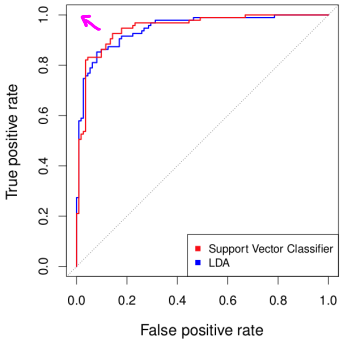
$e^{-\text{big}} \rightsquigarrow$ small

Radial kernels $\rightsquigarrow$ measure local behavior

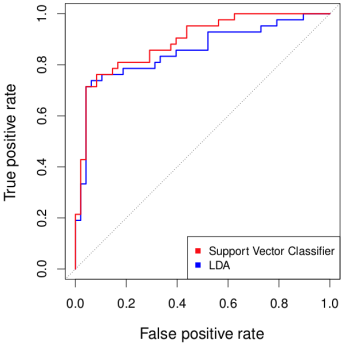


Example with heart data

SVC



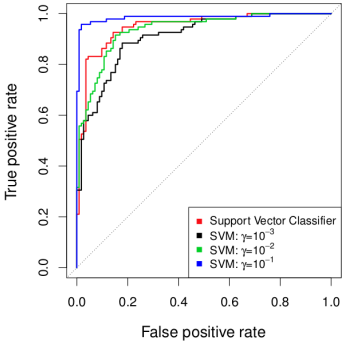
Training



Testing

Example with heart data

SVM



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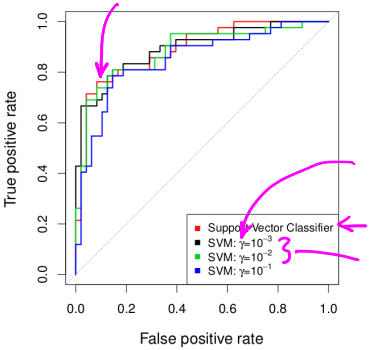

Testing

Section 3

SVM with more than two classes

One-Vs-One Classification

Also called all-pairs

- Want to predict $K > 2$ classes
- Construct $\binom{K}{2}$ SVMs, each one learning class k vs class k'

Apples
Bananas
Oranges
Peaches

Apples vs orange $\rightsquigarrow$ predict
Banana vs Apples $\rightsquigarrow$ prediction
oranges . . .
⋮

final prediction label w/ highest vote

One-Vs-All Classification

Want to predict $K > 2$ classes

Fit K SVMs $\rightsquigarrow$ get $f_k(x)$ as model

Predict k vs not k

apples $\rightsquigarrow$ 1 = apples -1 = anything else

oranges $\rightsquigarrow$ 1 = oranges -1 = anything else

bananas

Assign k for which $f_k(x)$ is largest

TL;DR

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i \overset{K(x, x_i)}{\cancel{\langle x, x_i \rangle}}$$

Kernels

- Linear

$$K(x_i, x_{i'}) = \sum_{j=1}^{p} x_{ij} x_{i'j}$$

- Polynomial

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^d$$

- Radial

$$K(x_i, x_i') = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right)$$

Next time

Next time:

- SVM Lab

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